AI-Based Plagiarism Detection v0.1

Report ID: plagcheck-20261015094221897822Z
Submission: srs.pdf
Analysis Timestamp: 2026-10-15T09:42:21.899292+00:00
Overall Similarity: 1.0

Methodology

The overall score is the highest direct-overlap coverage against any one reference. Exact passages are matched in order after Unicode normalization; near-exact passages may bridge small insertions or deletions only when the ordered sequence remains sufficiently similar. TF-IDF cosine similarity is shown as a separate contextual signal and does not increase the direct-overlap score. These signals are not proof of plagiarism.

Matching passages

Reference: srs.pdf | Type: exact | Similarity: 1.0
Submission excerpt: Software Requirement Specifications AI-Based Plagiarism Detection Framework Prepared for: Dr. Dheeraj Kumar Dixit Prepared by: Yug Jaiswal Version 1.0 1. Executive Summary This project proposes an AI-powered academic integrity framework that detects direct plagiarism, paraphrased plagiarism, cross-document similarity, and estimates the likelihood of AI-generated text. The system is designed as a decision-support tool for educators rather than a proof engine. 2. Problem Statement Existing plagiarism tools primarily detect copied text, while AI-generated assignments and advanced paraphrasing make academic integrity harder to assess. The framework combines textual similarity, semantic similarity, and stylometric AI analysis in one platform. 3. Project Objectives ■ Detect direct plagiarism using textual similarity. ■ Detect paraphrased plagiarism using semantic similarity. ■ Identify partial plagiarism at sentence/paragraph level. ■ Compare submissions across institutional documents. ■ Estimate AI-generated text likelihood using linguistic features. ■ Generate explainable reports for faculty. 4. Proposed System Modules Module Purpose Document Processing Extract text from TXT, PDF, DOCX and preprocess it. Plagiarism Engine TF-IDF and semantic similarity analysis. AI Text Analysis Stylometric and machine learning based AI likelihood estimation. Evidence Engine Highlight suspicious passages with confidence scores. Reporting Module Generate explainable reports and recommendations. Administration Module Manage datasets, users, and institutional repository. 5. Functional Requirements 1. Upload or paste academic documents. 2. Extract and preprocess text. 3. Detect copied content. 4. Detect paraphrased content. 5. Compare multiple documents. 6. Estimate AI-generated text likelihood. 7. Highlight suspicious passages. 8. Generate downloadable reports. 6. Non-Functional Requirements ■ Accuracy through multiple evidence sources. ■ Response time under practical academic document sizes. ■ Secure handling of uploaded documents. ■ Scalable architecture for institutional deployment. ■ Explainable output for human review. 7. Technology Stack Layer Technology Frontend React / HTML-CSS-JS Backend Python Flask or FastAPI NLP spaCy, NLTK, Sentence Transformers ML scikit-learn, Hugging Face Transformers Database PostgreSQL / MySQL Storage Docker 8. Research Components ■ Natural Language Processing ■ Semantic Similarity ■ Stylometry ■ Transformer Models ■ Explainable AI ■ Evaluation Metrics (Precision, Recall, F1, ROC-AUC) 9. Risks & Limitations ■ AI detection is probabilistic. ■ False positives and false negatives are possible. ■ Paraphrased AI text is harder to detect. ■ Similarity alone does not prove plagiarism. ■ Requires periodic retraining for new AI models. 10. Project Milestones Milestone’s Phase/Work Date Milestone 01 S.R.S. and Literature Review 21 August 2026 Milestone 02 Version 0.1 (Text/Document Preprocessing) 04 September 2026 Milestone 03 Version 0.2 (Plagiarism

Engine) 18 September 2026 Milestone 04 Version 0.3 (Ai Text Module) 02 October 2026 Milestone 05 Version 0.4 (Evidence & Reporting) 16 October 2026 Milestone 06 Integration and Project Review 30 October 2026 11. Deliverables ■ SRS document. ■ System architecture. ■ Prototype web application. ■ Trained AI detection model. ■ Evaluation report. ■ Presentation slides. ■ Final technical report. 12. Success Criteria The project is successful if it detects multiple forms of plagiarism, produces explainable reports, and evaluates AI-generated text using measurable metrics on benchmark datasets.

Reference excerpt: Software Requirement Specifications AI-Based Plagiarism Detection Framework Prepared for: Dr. Dheeraj Kumar Dixit Prepared by: Yug Jaiswal Version 1.0 1. Executive Summary This project proposes an AI-powered academic integrity framework that detects direct plagiarism, paraphrased plagiarism, cross-document similarity, and estimates the likelihood of AI-generated text. The system is designed as a decision-support tool for educators rather than a proof engine. 2. Problem Statement Existing plagiarism tools primarily detect copied text, while AI-generated assignments and advanced paraphrasing make academic integrity harder to assess. The framework combines textual similarity, semantic similarity, and stylometric AI analysis in one platform. 3. Project Objectives ■ Detect direct plagiarism using textual similarity. ■ Detect paraphrased plagiarism using semantic similarity. ■ Identify partial plagiarism at sentence/paragraph level. ■ Compare submissions across institutional documents. ■ Estimate AI-generated text likelihood using linguistic features. ■ Generate explainable reports for faculty. 4. Proposed System Modules Module Purpose Document Processing Extract text from TXT, PDF, DOCX and preprocess it. Plagiarism Engine TF-IDF and semantic similarity analysis. AI Text Analysis Stylometric and machine learning based AI likelihood estimation. Evidence Engine Highlight suspicious passages with confidence scores. Reporting Module Generate explainable reports and recommendations. Administration Module Manage datasets, users, and institutional repository. 5. Functional Requirements 1. Upload or paste academic documents. 2. Extract and preprocess text. 3. Detect copied content. 4. Detect paraphrased content. 5. Compare multiple documents. 6. Estimate AI-generated text likelihood. 7. Highlight suspicious passages. 8. Generate downloadable reports. 6. Non-Functional Requirements ■ Accuracy through multiple evidence sources. ■ Response time under practical academic document sizes. ■ Secure handling of uploaded documents. ■ Scalable architecture for institutional deployment. ■ Explainable output for human review. 7. Technology Stack Layer Technology Frontend React / HTML-CSS-JS Backend Python Flask or FastAPI NLP spaCy, NLTK, Sentence Transformers ML scikit-learn, Hugging Face Transformers Database PostgreSQL / MySQL Storage Docker 8. Research Components ■ Natural Language Processing ■ Semantic Similarity ■ Stylometry ■ Transformer Models ■ Explainable AI ■ Evaluation Metrics (Precision, Recall, F1, ROC-AUC) 9. Risks & Limitations ■ AI detection is probabilistic. ■ False positives and false negatives are possible. ■ Paraphrased AI text is harder to detect. ■ Similarity alone does not prove plagiarism. ■ Requires periodic retraining for new AI models. 10. Project Milestones Milestone’s Phase/Work Date Milestone 01 S.R.S. and Literature Review 21 August 2026 Milestone 02 Version 0.1 (Text/Document Preprocessing) 04 September 2026 Milestone 03 Version 0.2 (Plagiarism Engine) 18 September 2026 Milestone 04 Version 0.3 (Ai Text Module) 02 October 2026 Milestone 05 Version 0.4 (Evidence & Reporting) 16 October 2026 Milestone 06 Integration and Project Review 30 October 2026 11. Deliverables ■ SRS document. ■ System architecture. ■ Prototype web application. ■ Trained AI detection model. ■ Evaluation report. ■ Presentation slides. ■ Final technical report. 12. Success Criteria The project is successful if it detects multiple forms of plagiarism, produces explainable reports, and evaluates AI-generated text using measurable metrics on benchmark datasets.